In[11]:=
```
$Assumptions = {r > 0, ϕ ∈ Reals, σ ∈ Reals, R > 0, S > 0}
M = 1;
```
Out[11]=

$\{r > 0, \phi \in \mathbb{R}, \sigma \in \mathbb{R}, R > 0, S > 0\}$

Ortsvektor, Geschwindigkeit der Akkretionsscheibe und Drehimpuls des Schwarzen Lochs

In[19]:=
```
rv[r_, ϕ_] = r * {Cos[ϕ], Sin[ϕ], 0}
vv[r_, ϕ_] = Sqrt[M / r] * {-Sin[ϕ], Cos[ϕ], 0}
Sv[ϵ_, S_] = S * {Sin[ϵ], 0, Cos[ϵ]}
```
Out[19]=

$\{r \operatorname{Cos}[\phi], r \operatorname{Sin}[\phi], 0\}$

Out[20]=

$\left\{-\sqrt{\frac{1}{r}} \operatorname{Sin}[\phi], \sqrt{\frac{1}{r}} \operatorname{Cos}[\phi], 0\right\}$

Out[21]=

$\{S \operatorname{Sin}[\epsilon], 0, S \operatorname{Cos}[\epsilon]\}$

Drehmoment

In[23]:=
```
rv[r, ϕ]×(σ / r ^ 2 * (2 * vv[r, ϕ]×Sv[ϵ, S] - 6 * Sv[ϵ, S].rv[r, ϕ] / r ^ 2 * vv[r, ϕ]×rv[r, ϕ])) // FullSimplify
```
Out[23]=

$\left\{\frac{2 S \sigma \operatorname{Sin}[\epsilon] \operatorname{Sin}[2 \phi]}{r^{3/2}}, -\frac{4 S \sigma \operatorname{Cos}[\phi]^2 \operatorname{Sin}[\epsilon]}{r^{3/2}}, 0\right\}$

In[38]:=

$Mv = \operatorname{Integrate}\left[\left\{\frac{2 S \sigma \operatorname{Sin}[\epsilon] \operatorname{Sin}[2 \phi]}{r^{3/2}}, -\frac{4 S \sigma \operatorname{Cos}[\phi]^2 \operatorname{Sin}[\epsilon]}{r^{3/2}}, 0\right\}, \{r, 1, R\}, \{\phi, 0, 2\pi\}\right] // \operatorname{FullSimplify}$

Out[38]=

$\left\{0, 8\pi\left(-1+\frac{1}{\sqrt{R}}\right) S \sigma \operatorname{Sin}[\epsilon], 0\right\}$

Drehimpuls

In[28]:=
```
σ * r * rv[r, ϕ]×vv[r, ϕ] // FullSimplify
```
Out[28]=

$\{0, 0, r^{3/2} \sigma\}$

In[37]:=

$Lv = \operatorname{Integrate}\left[\{0, 0, r^{3/2} \sigma\}, \{r, 1, R\}, \{\phi, 0, 2\pi\}\right] // \operatorname{FullSimplify}$

Out[37]=

$\left\{0, 0, \frac{4}{5} \pi\left(-1+R^{5/2}\right) \sigma\right\}$

Präzession

In[35]:= $\text{Solve}\left[\left\{0,\ 8\,\pi\left(-1+\frac{1}{\sqrt{R}}\right)S\,\sigma\,\text{Sin}[\epsilon],\ 0\right\} == \left\{0,\ 0,\ \frac{4}{5}\,\pi\left(-1+R^{5/2}\right)\sigma\right\}*\text{Sv}[\epsilon,\ \Omega],\ \Omega\right]$

Out[35]=

$$\left\{\left\{\Omega\to-\frac{10\,S}{\sqrt{R}\left(1+\sqrt{R}+R+R^{3/2}+R^2\right)}\right\}\right\}$$

In[44]:= $-\frac{10\,S}{\sqrt{R}\left(1+\sqrt{R}+R+R^{3/2}+R^2\right)}$ // FullSimplify

Out[44]=

$$-\frac{10\,S}{\sqrt{R}+R+R^{3/2}+R^2+R^{5/2}}$$

In[43]:= $\Omega v = \text{Sv}\left[\epsilon,\ -\frac{10\,S}{\sqrt{R}+R+R^{3/2}+R^2+R^{5/2}}\right]$ // FullSimplify

Out[43]=

$$\left\{-\frac{10\,S\,\text{Sin}[\epsilon]}{\sqrt{R}+R+R^{3/2}+R^2+R^{5/2}},\ 0,\ -\frac{10\,S\,\text{Cos}[\epsilon]}{\sqrt{R}+R+R^{3/2}+R^2+R^{5/2}}\right\}$$